Chapter

Hydrogen Chloride

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1. A salt which is soluble in hot water, but insoluble in cold water. SQP 2023

(a) $PbCl_2$
(b) $Pb(NO_3)_2$
(c) HClO
(d) KCl

2. Name the experiment which demonstrates that hydrogen chloride is very soluble in water. MAIN 2023

(a) Fountain experiment
(b) Catalytic reduction of hydrogen chloride
(c) Catalytic oxidation of ammonia
(d) Bosch process

3. A compound which on reaction with HCl leaves a yellow residue behind. SQP 2021

(a) AgCl
(b) $Na_2S_2O_3$
(c) Conc. H_2SO_4
(d) MnO_2

4. .... combines with unreactive metals like gold a, platinum. MAIN 2021

(a) Nascent Oxygen [O]
(b) Nascent Iodine [I]
(c) Nascent Nitrogen [O]
(d) Nascent Chlorine [Cl]

5. Assertion : Hydrogen chloride gas does not conduct electricity.
Reason : Hydrogen chloride gas is a covalent compound. **COMP 2022**

(a) Both assertion and reason are true and reason is the correct explanation of assertion.
(b) Both assertion and reason are true but reason is not the correct explanation of Assertion.
(c) Assertion is true but reason is false.
(d) Assertion is false but reason is true.

6. Two compounds of Lead which combine with conc. HCl to liberate Chlorine. COMP 2021

7. Solvent for noble metal. SQP 2022

8. Assertion : Dry hydrogen chloride gas is collected by the upward displacement of air.
Reason : Hydrogen chloride gas is lighter than air. **SQP 2025**

(a) Both assertion and reason are true and reason is the correct explanation of assertion.
(b) Both assertion and reason are true but reason is not the correct explanation of assertion.
(c) Assertion is true but reason is false.
(d) Assertion is false but reason is true.

9. Hydrogen chloride and water are examples of_____(polar covalent compounds, nonpolar covalent compounds) and a solution of hydrogen chloride in water_____. (contains, does not contain) free ions. SQP 2023

10. $HNO_3 + HCl \longrightarrow$ _____ + _____ + ______. **MAIN 2023**

11. $NaOH + HCl \longrightarrow$ _____ + _____. **MAIN 2023**

12. What property of hydrogen chloride is demonstrated when it is collected by downward delivery (upward displacement)? SQP 2023

13. Outline the steps required to convert hydrogen chloride to anhydrous iron(III) chloride. Write the equations for the reactions which take place. MAIN 2023

14. The following questions are related to the laboratory preparation of hydrochloride acid.

(i) Name the reactants used.

(ii) Give suitable reason for choosing these reactants.

(iii) Write balanced chemical equation for the reaction taking place below 200°C.

(iv) Why the temperature of the reaction must be maintained below 200°C ?

Solution

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1. (a) $PbCl_2$

Lead chloride ($PbCl_2$) is a salt that is **insoluble in cold water** but **soluble in hot water**. This is due to the **borderline acidic nature of the lead ion**, as per the **hard-soft acid-base (HSAB) theory**, which affects its solubility.

2. (a) Fountain experiment

The **fountain experiment** demonstrates that **hydrogen chloride (HCl) gas is highly soluble in water**. When HCl gas comes into contact with water in the experiment, it **dissolves immediately**, creating a visible "fountain" effect due to the rapid movement of water into the gas container.

3. (c) Conc. H_2SO_4

When **concentrated sulfuric acid (H_2SO_4)** reacts with certain substances, it can leave a **yellow residue**, indicating the formation of compounds like **sulfur or sulfur-containing products**.

4. (d) Nascent Chlorine [Cl]

Nascent chlorine is highly reactive and can combine with **unreactive metals** such as **gold (Au) and platinum (Pt)**, which normally do not react with ordinary chlorine. This property makes nascent chlorine useful in certain chemical processes.

5. (a) Both assertion and reason are true and reason is the correct explanation of assertion

Hydrogen chloride (HCl) is a **covalent compound** in its gaseous state. Since it **does not contain free ions**, it **cannot conduct electricity**. Therefore, the assertion is true, and the reason correctly explains why HCl gas is non-conductive.

6. Lead dioxide and Red lead (Pb_3O_4)

7. Aqua regia.

8. (c) Assertion is true but reason is false

Dry hydrogen chloride (HCl) gas is collected by **upward displacement of air** because it is **denser than air**, not lighter. The reason given is incorrect, but the assertion about the method of collection is true.

9. polar covalent compounds, cotains.

10. $HNO_3 + 3HCl \longrightarrow NOCl + 2[Cl] + 2H_2O$

11. $NaOH + HCl \longrightarrow NaCl + H_2O$

12. The property of **hydrogen chloride (HCl)** demonstrated when it is collected by **downward delivery (upward displacement of air)** is that **HCl gas is heavier than air**. This allows it to **displace air from below** and collect at the bottom of the container.

13. To convert **hydrogen chloride (HCl)** to **anhydrous iron(III) chloride ($FeCl_3$)**, the following steps are followed:

1. **Preparation of chlorine:** React concentrated HCl with an oxidizing agent like MnO_2, PbO_2, or Pb_3O_4.

$$4HCl + MnO_2 \longrightarrow Cl_2 + 2H_2O + MnCl_2$$

1. **Drying of chlorine:** Pass the chlorine gas through **concentrated H_2SO_4** to remove moisture.
2. **Formation of anhydrous $FeCl_3$:** Pass **dry chlorine gas** over **heated iron**:

$$2Fe + 3Cl_2 \xrightarrow{\text{heated, dry}} 2FeCl_3$$

This gives **anhydrous iron(III) chloride**.

14. (i) Reactants used:
Sodium chloride (NaCl) and concentrated sulphuric acid (H_2SO_4).

(ii) Reason for choosing these reactants:
Sodium chloride is **easily and cheaply available**, and concentrated sulphuric

acid is chosen because it is the **least volatile acid**, making the reaction safer and more efficient.

(iii) Balanced chemical equation below 200°C:

$$\mathrm{NaCl} + \mathrm{H_2SO_4} \xrightarrow{<200^\circ\mathrm{C}} \mathrm{NaHSO_4} + \mathrm{HCl}\uparrow$$

(iv) Reason for maintaining temperature below 200°C:

- If heated too much, **excess fuel is wasted**.
- Higher temperatures produce a **sticky mass of sodium sulphate**, which can **damage the glass apparatus**.